12.5 Critical concentration and toxicity of minerals

12.5.1 Critical Concentration

To increase the productivity and also to avoid mineral toxicity knowledge of critical concentration is essential. Mineral nutrients lesser than critical concentration cause deficiency symptoms. Increase of mineral nutrients more than the normal concentration causes toxicity. A concentration, at which 10 % of the dry weight of tissue is reduced, is considered as toxic. Figure 12.2 explains about Critical Concentration.

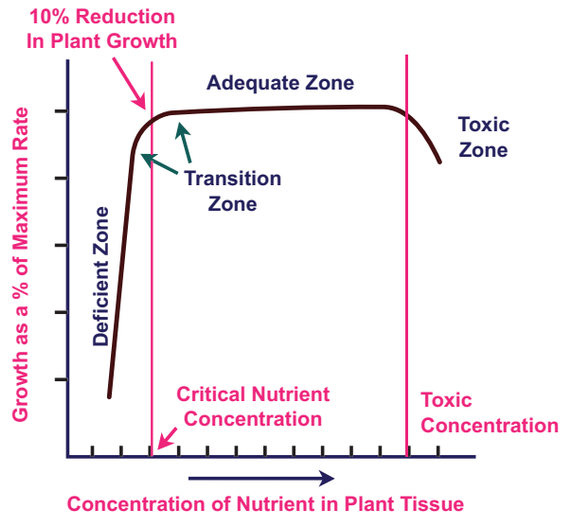


Figure 12.2: Critical Concentration

12.5.2 Mineral Toxicity

a. Manganese toxicity

Increased Concentration of Manganese will prevent the uptake of Fe and Mg, prevent translocation of Ca to the shoot apex and cause their deficiency. The symptoms of manganese toxicity are appearance of brown spots surrounded by chlorotic veins.

b. Aluminium Toxicity

Aluminium toxicity causes precipitation of nucleic acid, inhibition of ATPase, inhibition of cell division and binding of plasma membrane with Calmodulin.

For theories regarding, translocation of minerals please refer Chapter- 11.

12.6 Hydroponics and Aeroponics

1. Hydroponics or Soilless culture: Von Sachs developed a method of growing plants in nutrient solution. The commonly used nutrient solutions are **Knop solution** (1865) and **Arnon** and **Hoagland Solution** (1940). Later the term Hydroponics was coined by **Goerick** (1940) and he also introduced commercial techniques for hydroponics. In hydroponics roots are immersed in the solution containing nutrients and air is supplied with help of tube (Figure 12.3).

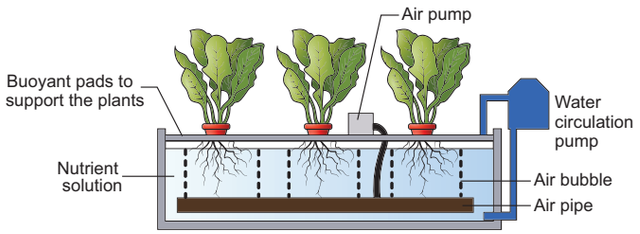


Figure 12.3: Hydroponics

Aeroponics: This technique was developed by **Soifer Hillel** and **David Durger**. It is a system where roots are suspended in air and nutrients are sprayed over the roots by a motor driven rotor (Figure 12.4).

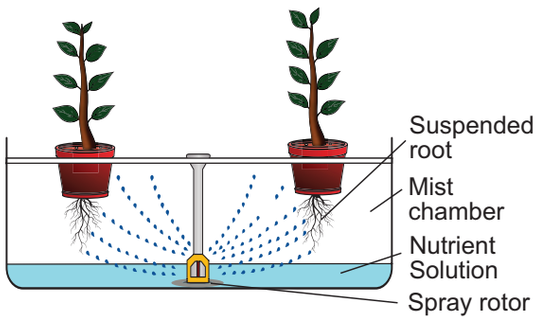


Figure 12.4: Aeroponics

12.7 Nitrogen Fixation

Inspiring act of nature is self-regulation. As all living organisms act as tools for biogeochemical cycles, nitrogen cycle is highly regulated. Life on earth depends on nitrogen cycle. Nitrogen occurs in atmosphere in the form of N_2 (N≡N), two nitrogen atoms joined together by strong triple covalent bonds. The process of converting atmospheric nitrogen (N_2) into ammonia is termed as nitrogen fixation. Nitrogen fixation

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